

CS773 Checkpoint 1

CacheBusters:

Detecting Cache-based Covert Channel Attacks Using Cache Miss and Access Patterns

Suraj Patil, Shubh Nagar, Abhishek Singh
CacheBusters

24m0837@iitb.ac.in, shubh.nagar@cse.iitb.ac.in, 24m0808@iitb.ac.in



Perf tool

- Powerful CLI tool for analyzing CPU usage, cache misses, and performance bottlenecks in applications and the kernel
- Processors are equipped with registers that keep tracks of CPU related events such as cache misses, branch misses etc.
- These registers are called HPCs(Hardware Performance Counters)

Perf Drawback

- Perf -stat command used to get stats/samples
- The minimum interval between two consecutive samples is 100 ms
- This interval is much less than certain types of attack

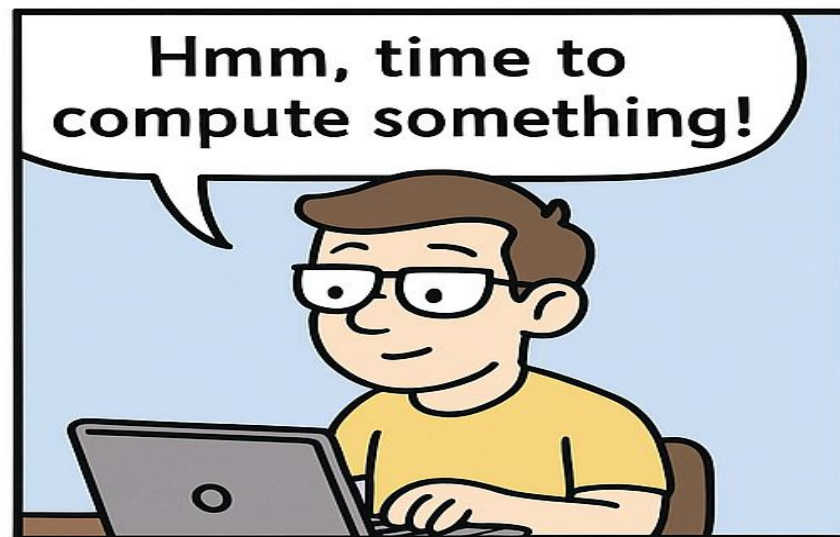
Quickhpc

- Library used for probing HPCs is PAPI (Performance Application Programming Interface)
- PAPI is a standardized library interface that provides access to hardware performance counters found in most modern CPUs
- main reason quickhpc uses PAPI is its high resolution(time between two samples)
- Running quickhpc the required arguments are the PID of the process to monitor and the list of events to be monitored

Method used for Flush+Reload

- Correlation Based
 - Finding a correlation between 2 processes analyzing the data collected by quickhpc
 - Number of total L3 cache accesses over time as a good indicator of correlation
 - Analyse the differences between the 2 processes data dump

- Since each iteration accesses the same number of memory addresses
- Total L3 cache accesses over time serve as a good correlation indicator for both attacks
- We also discussed how we can detect anomalies, but it's hard using just correlation but we will explore more ideas on it.



Checkpoint 1

- Implement cache-based covert channel Attack - Prime+Probe.
- Analyze cache miss data for flush+reload attack using perf tools(quickhpc).
- Measure timing variations using RDTSCP to identify access delays.
- Identify cache miss patterns over time to detect periodic behavior.
- Apply correlation analysis to identify covert communication signals.

Plan for checkpoint-II

- Analyse cache miss patterns for prime+probe attack.
- Identify false positives and false negatives in detection results.
- Optimize cache miss threshold and autocorrelation analysis for better detection.
- Introduce noise to reduce effectiveness of the attack and analyse.
- Measure the impact of noise on attack success rate and data leakage.
- Enhance countermeasures for better efficiency by dynamically adjusting noise injection based on cache miss patterns and attack success rates.

Github link

- <https://github.com/shubhnagar/CS773/>